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ME630
Assignment - 3

We have 1D Burger's equation

$$\frac{\partial v}{\partial t} + v\frac{\partial v}{\partial x} = \nu \frac{\partial^2 v}{\partial x^2} \qquad \nu = \frac{1}{Re}$$

Now descritize it into explicitly, implicitly and by crank Nicolson method -

Explicit method -

$$\frac{\partial v}{\partial t} + \frac{\partial (v^2/2)}{\partial x} = \nu \frac{\partial^2 v}{\partial x^2}$$

$$\nu = \frac{1}{Re} \quad \text{and} \quad F = \frac{v^2}{2}$$

$$\frac{\partial v}{\partial t} + \frac{\partial (F)}{\partial x} = \frac{1}{Re} \frac{\partial^2 v}{\partial x^2}$$

Consider forward propagating wave $v > 0$ so we use upwind scheme

$$\frac{v_j^{n+1} - v_j^n}{\Delta t} + \frac{F(v_j^n) - F(v_{j-1}^n)}{\Delta x} = \frac{1}{Re} \frac{1}{\Delta x^2} (v_{j+1}^n - 2v_j^n + v_{j-1}^n)$$

$$v_j^{n+1} - v_j^n = -\frac{\Delta t}{\Delta x}\left[F(v_j^n) - F(v_{j-1}^n)\right] + \frac{\Delta t}{Re \Delta x^2}\left(v_{j+1}^n - 2v_j^n + v_{j-1}^n\right)$$

$$\Rightarrow v_j^{n+1} = v_j^n - \frac{\Delta t}{\Delta x}\left[F(v_j^n) - F(v_{j-1}^n)\right] + \frac{\Delta t}{Re \Delta x^2}\left(v_{j+1}^n - 2v_j^n + v_{j-1}^n\right)$$

So LHS is unknown v_j^{n+1} and RHS is known terms - so it is directly solved.

Implicit method —

$$\frac{\partial v}{\partial t} + \frac{\partial (F)}{\partial x} = \nu \frac{\partial^2 v}{\partial x^2} \qquad \nu = \frac{1}{Re}$$

$$\frac{\Delta v_j^{n+1}}{\Delta t} + Lx\left(F_j^{n+1}\right) = \nu\, Lxx\left(v_j^{n+1}\right) \qquad -(A)$$

where $\Delta v_j^{n+1} = v_j^{n+1} - v_j^n$

$$Lx\left(v_j^{n+1}\right) = \left(v_{j+1}^{n+1} - v_{j-1}^{n+1}\right)/2\Delta x$$

$$Lxx\left(v_j^{n+1}\right) = \left(\frac{v_{j+1}^{n+1} - 2v_j^{n+1} + v_{j-1}^{n+1}}{\Delta x^2}\right)$$

By Taylor's ~~root~~ Series

$$F_j^{n+1} = F_j^n + \Delta t \left.\frac{\partial F}{\partial t}\right|_j^n + \frac{\Delta t^2}{2!} \left.\frac{\partial^2 F}{\partial t^2}\right|_j^n$$

$$\frac{\partial F}{\partial t} = v \frac{\partial v}{\partial t}$$

$$\Rightarrow F_j^{n+1} = F_j^n + \Delta t\, v \frac{\partial v}{\partial t} + \text{Higher order terms}$$

$$\Rightarrow F_j^{n+1} = F_j^n + v_j^n \left(v_j^{n+1} - v_j^n\right)$$

$$\Rightarrow F_j^{n+1} = v_j^n v_j^{n+1} - 0.5\, v_j^n v_j^n$$

Now put in main equation — (A)

$$\frac{u_j^{n+1}-u_j^n}{\Delta t} + L_x\left(u_j^n u_j^{n+1} - 0.5 u_j^n u_j^n\right) = \nu L_{xx}\left(u_j^{n+1}\right)$$

$$\frac{u_j^{n+1}-u_j^n}{\Delta t} + \frac{u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1}}{2\Delta x} = \nu L_{xx}\left(u_j^{n+1}\right) + 0.5 L_x\left(u_j^n u_j^n\right)$$

$$\frac{u_j^{n+1}-u_j^n}{\Delta t} + \frac{u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1}}{2\Delta x} - \nu\left(\frac{u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}}{\Delta x^2}\right) = 0.5\left(\frac{u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n}{2\Delta x}\right)$$

$$u_j^{n+1} - u_j^n + \frac{\Delta t}{2\Delta x}\left(u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1}\right) - \frac{\nu\Delta t}{\Delta x^2}\left(u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}\right) = 0.5\frac{\Delta t}{2\Delta x}\left(u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n\right)$$

$$2\Delta x^2 u_j^{n+1} - 2\Delta x^2 u_j^n + \Delta t\Delta x\left(u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1}\right) - 2\nu\Delta t\left(u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}\right) = 0.5\Delta t\Delta x\left(u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n\right)$$

By rearranging

RHS → all known term

LHS → all unknown terms.

$$u_{j-1}^{n+1}\left[-\Delta t\Delta x\, u_{j-1}^{n} - 2\nu\Delta t\right] + u_j^{n+1}\left[4\nu\Delta t + 2\Delta x^2\right] + u_{j+1}^{n+1}\left[\Delta t\Delta x\, u_{j+1}^{n} - 2\nu\Delta t\right] = 2\Delta x^2 u_j^n + 0.5\Delta x\Delta t\left[u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n\right]$$

where $\nu = \frac{1}{Re}$

$$u_{j-1}^{n+1}\left[-\Delta t\Delta x\, u_{j-1}^{n} - \frac{2\Delta t}{Re}\right] + u_j^{n+1}\left[\frac{4\Delta t}{Re} + 2\Delta x^2\right] + u_{j+1}^{n+1}\left[\Delta t\Delta x\, u_{j+1}^{n} - \frac{2\Delta t}{Re}\right] = 2\Delta x^2 u_j^n + 0.5\Delta x\Delta t\left[u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n\right]$$

$$a = -\Delta t\Delta x\, u_{j-1}^n - \frac{2\Delta t}{Re}$$

$$b = \frac{4\Delta t}{Re} + 2\Delta x^2$$

$$c = \Delta t\Delta x\, u_{j+1}^n - \frac{2\Delta t}{Re}$$

$$d = 2\Delta x^2 u_j^n + 0.5\Delta x\Delta t\left[u_{j+1}^n u_{j+1}^n - u_{j-1}^n u_{j-1}^n\right]$$

$$a\, u_{j-1}^{n+1} + b\, u_j^{n+1} + c\, u_{j+1}^{n+1} = d$$

In this we can apply thomas algorith to find u_{j-1}^{n+1}, u_j^{n+1}, u_{j+1}^{n+1}

Crank Nicolson Method –

$$\frac{\partial u}{\partial t} + \frac{\partial F}{\partial x} = \nu \frac{\partial^2 v}{\partial x^2} \qquad , \nu = \frac{1}{Re}$$

Similarly

$$\Rightarrow \frac{\Delta v_j^{n+1}}{\Delta t} + 0.5\, Lx\left(E_j^n + F_j^{n+1}\right) = 0.5\nu\, Lxx\left[v_j^{n+1} + v_j^n\right] \quad (B)$$

where $Lx(v_j^n) = \dfrac{v_{j+1}^n - v_{j-1}^n}{2\Delta x}$

$$Lxx(v_j^n) = \frac{v_{j+1}^n - 2v_j^n + v_{j-1}^n}{\Delta x^2}$$

By Taylor's Series

$$F_j^{n+1} = F_j^n + \Delta t \left.\frac{\partial F}{\partial t}\right|_j^n + \text{higher order term}$$

$$\frac{\partial F}{\partial t} = v\frac{\partial v}{\partial t}$$

$$F_j^{n+1} = F_j^n + \Delta t\, v \frac{\partial v}{\partial t}$$

$$F_j^{n+1} = F_j^n + v_j^n \Delta v_j^{n+1}$$

$$\Rightarrow F_j^{n+1} = F_j^n + v_j^n\left(v_j^{n+1} - v_j^n\right)$$

$$\Rightarrow F_j^{n+1} = F_j^n + v_j^n v_j^{n+1} - v_j^n v_j^n$$

Put in main equation (B)

$$\frac{v_j^{n+1} - v_j^n}{\Delta t} + 0.5\, Lx\left(2F_j^n + v_j^n v_j^{n+1} - 2F_j^n\right) = 0.5\nu\, Lxx\left[v_j^n + v_j^{n+1}\right]$$

$$u_j^{n+1} - u_j^n + 0.5\Delta t L_x(u_j^n u_j^{n+1}) = 0.5\nu\Delta t L_{xx}(u_j^n + u_j^{n+1})$$

$$u_j^{n+1} + 0.5\Delta t L_x(u_j^n u_j^{n+1}) - 0.5\nu\Delta t L_{xx}(u_j^{n+1}) = u_j^n + 0.5\nu\Delta t L_{xx}(u_j^n)$$

$$u_j^{n+1} + 0.5\Delta t \frac{(u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1})}{2\Delta x} - 0.5\nu\Delta t \frac{(u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1})}{\Delta x^2}.$$

$$= u_j^n + 0.5\nu\Delta t \frac{(u_{j+1}^n - 2u_j^n + u_{j-1}^n)}{\Delta x^2}$$

By ~~rearranging we get~~

$$2\Delta x^2 u_j^{n+1} + 0.5\Delta t \Delta x (u_{j+1}^n u_{j+1}^{n+1} - u_{j-1}^n u_{j-1}^{n+1}) - \nu\Delta t (u_{j+1}^{n+1} - 2u_j^{n+1} + u_{j-1}^{n+1}) = 2\Delta x^2 u_j^n + \nu\Delta t (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

By rearranging

$$u_{j-1}^{n+1}\left[-0.5\Delta t\Delta x\, u_{j-1}^n - \nu\Delta t\right] + (2\Delta x^2 + 2\nu\Delta t) u_j^{n+1} + u_{j+1}^{n+1}\left[0.5\Delta t\Delta x\, u_{j+1}^n - \nu\Delta t\right] = 2\Delta x^2 u_j^n + \nu\Delta t (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

$$a = -0.5\Delta t\Delta x\, u_{j-1}^n - \nu\Delta t$$

$$b = 2\Delta x^2 + 2\nu\Delta t$$

$$c = 0.5\Delta t\Delta x\, u_{j+1}^n - \nu\Delta t$$

$$d = 2\Delta x^2 u_j^n + \nu\Delta t (u_{j+1}^n - 2u_j^n + u_{j-1}^n)$$

$$\boxed{a u_{j-1}^{n+1} + b u_j^{n+1} + c u_{j+1}^{n+1} = d}$$

In this we can apply Thomas algorithm to find u_{j-1}^{n+1}, u_j^{n+1}, u_{j+1}^{n+1}